

Table S2. Definition of outcomes

Outcomes	Definition	HOPE studies	NMP studies
Major complication	Clavien-Dindo ^{15,16} classification ≥ 3	Rest of the studies	Bral 2017, Mergental 2020, Gaurav 2022, Wehrle 2024
	Clavien-Dindo ^{15,16} classification $\geq 3b$	Ravaioli 2022	Selzner 2016, Nasralla 2018, Fodor 2021, Hefler 2023
EAD	Presence of ≥ 1 of: postoperative liver injury indicated by bilirubin ≥ 10 mg/dL or INR >1.6 on day 7, or ALT/AST >2000 IU/L in the first 7 days ¹⁷	All studies	Rest of the studies
	No definition provided	-	Bral 2017
NAS	Stricture, dilation, or irregularity of intra- or extrahepatic bile ducts on MRCP, excluding isolated anastomotic strictures in the presence of patent arterial vasculature	Dutkowski 2015, Patrono 2019, van Rijn 2021	Fodor 2021, Gaurav 2022, Hefler 2023
	MRCP at six months showing stricture $>70\%$ of luminal diameter	-	Nasralla 2018
	Ischemic-type biliary lesion was reported, which was defined as any donor biliary tract with nonanastomotic stenosis requiring an endoscopic or radiological procedure in the absence of vascular complications	-	Ghinolfi 2019
	Endoscopic ductular narrowing with upstream dilation occurring at nonanastomotic locations	Panayotova 2024	-

Outcomes	Definition	HOPE studies	NMP studies
	Ischemic biliary complications (combination of non-anastomotic strictures and bile leaks)	-	Markmann 2022
	Any biliary stricture superior/proximal to the biliary anastomosis	-	Wehrle 2024
	No definition provided	Rest of the studies	Liu 2020, Mergental 2020, Chapman 2023
PNF¹⁸	Irreversible graft dysfunction leading to death or emergency retransplant within 7 days post-surgery, without immunologic or surgical causes	van Rijn 2017, van Rijn 2021	Markmann 2022, Gaurav 2022
	Patient death or the need of retransplantation within the first seven postoperative days excluding acute vascular complications	Ravaioli 2022	-
	Liver failure requiring retransplantation or leading to death within 7 days after transplantation	Rayar 2021	-
	Irreversible graft dysfunction, for non-technical and non-immunological causes, leading to death or emergency liver replacement during the first 10 days after liver transplantation	-	Nasralla 2018, Mergental 2020
	Meeting UNOS criteria for status 1A relisting and resulting in death or retransplant	Panayotova 2024	-
	No definition provided	Rest of the studies	Rest of the studies
Acute cellular rejection	Acute biopsy proven rejection	Jassem 2019, Schlegel 2023	Nasralla 2018
	Treated acute rejection	Schlegel 2019	-
	Requiring biopsy or medication	van Rijn 2021	-

Outcomes	Definition	HOPE studies	NMP studies
	Rejection activity index > 4	Dutkowski 2015	-
	According to Banff criteria ¹⁹	Panayotova 2024	-
	Histological evidence of rejection recorded within the first 6-month after transplantation on clinical suspicion	-	Gaurav 2022
	No definition provided	Patrono 2019, Ravaoli 2020, Ravaoli 2022, Czigany 2021	Hefler 2023, Wehrle 2024
AKI	Within 7 days, based on RIFLE criteria (≥ 2 x creatinine increase or >50% decrease in GFR)	Panayotova 2024	-
	Grade 2-3 AKI	Patrono 2019, Norén 2023	-
	Within 7 days, based on RIFLE criteria (≥ 2 x creatinine increase or >50% decrease in GFR), or the need for renal replacement therapy	-	Gaurav 2022
	Within 2 days, diagnosed according to KDIGO criteria	Rayar 2021	-
	No definition provided	-	Fodor 2021
PRS	A decrease in mean arterial pressure by at least 30% within 5 minutes after reperfusion, lasting for at least 1 minute ^{20,21}	Rayar 2021, Grąt 2023	Bral 2017, Nasralla 2018, Ghinolfi 2019, Chapman 2023
	A decrease in mean arterial pressure by at least 30% within 5 minutes after reperfusion, lasting for at least 1 minute (assessed in the context of inotrope use)	-	Mergental 2020

Outcomes	Definition	HOPE studies	NMP studies
	A drop exceeding 30% in mean systemic arterial blood pressure within 1 minute following reperfusion, regardless of whether the norepinephrine dose was doubled	van Rijn 2021	-
	A persistent drop in arterial blood pressure (>30% of the anhepatic level), asystole or hemodinamically significant arrhythmias	Patrono 2019	-
	A decrease of more than 30% of in the mean arterial blood pressure or the need for aminic support to maintain hemodynamic stability	Ravaioli 2022	-

HOPE, hypothermic oxygenated perfusion; NMP, normothermic machine perfusion; EAD, early allograft dysfunction; NAS, non-anastomotic biliary stricture; PNF, primary non-function; AKI, acute kidney injury; PRS, post-reperfusion syndrome; MRCP, magnetic resonance cholangiopancreatography; -, not applicable